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Course: Advance Programming

Write a C program for three different operations to analyze the space complexity, namely constant space $O(1)$, linear space $O(n)$, and quadratic space $O(n^2)$. Increase the input size and observe the memory used for each operation.

```
main.c
1  #include <stdio.h>
2  #include <stdlib.h>
3
4  void constantSpace()
5  {
6      int a = 5, b = 10, c;
7      c = a + b;
8  }
9
10 void linearSpace(int n)
11 {
12     int *arr = (int*)malloc(n * sizeof(int));
13
14     for(int i = 0; i < n; i++)
15         arr[i] = i;
16
17     free(arr);
18 }
19
20 void quadraticSpace(int n)
21 {
22     int **mat = (int**)malloc(n * sizeof(int*));
23
24     for(int i = 0; i < n; i++)
25         mat[i] = (int*)malloc(n * sizeof(int));
26
27     for(int i = 0; i < n; i++)
28         free(mat[i]);
29
30     free(mat);
31 }
32
33 int main()
34 {
35     int n;
36
37     printf("Enter input size: ");
38     scanf("%d", &n);
39
40     printf("\n--- Space Complexity Analysis ---\n");
41
42     printf("\nO(1) Constant Space Used: %lu bytes",
43         (unsigned long)(3 * sizeof(int)));
44
45     printf("\nO(n) Linear Space Used: %lu bytes",
46         (unsigned long)(n * sizeof(int)));
47
48     printf("\nO(n^2) Quadratic Space Used: %lu bytes\n",
49         (unsigned long)(n * n * sizeof(int)));
50
51     constantSpace();
52     linearSpace(n);
53     quadraticSpace(n);
54
55     return 0;
56 }
```

Output:

Output

```
Enter input size: 100
```

```
--- Space Complexity Analysis ---
```

```
O(1) Constant Space Used: 12 bytes
```

```
O(n) Linear Space Used: 400 bytes
```

```
O(n^2) Quadratic Space Used: 40000 bytes
```

```
=== Code Execution Successful ===
```